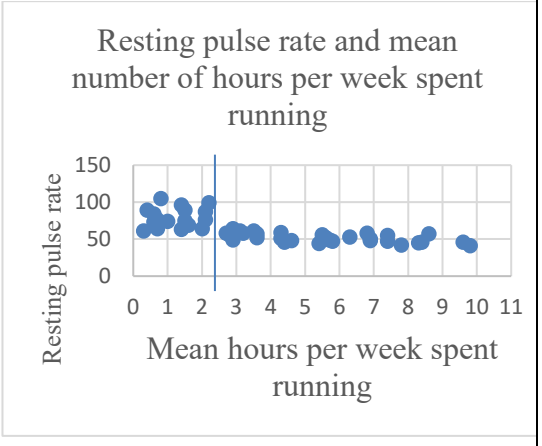
Question			Answer	Marks	AO	Guidance
15	(a)	(i)	opportunity (sampling)	B1 [1]	2.5	allow convenience (sampling) do not allow eg convenient sampling; eg opportunistic sampling
15	(a)	(ii)	only people known to the personal trainer / his clients sampled **oe** **so** eg they are fitter than typical members of the population **oe** eg sample may not be representative **oe**	B1 [1]	2.3	allow **so** any eg which alludes to sample possibly giving unrepresentative values
15	(b)		negative (association)	B1 [1]	2.4	ignore adjectives such as weak, strong etc **allow** eg as mean hours spent running increases, resting pulse rate decreases; eg negative relationship; **do not allow** eg (negative) **correlation**; eg as pulse rate increases, mean hours per week spent running decreases

Question			Answer	Marks	AO	Guidance
15	(c)			B1 [1]	2.2b	approximately vertical line at about 2.5; **B0** if more than one line

Question			Answer	Marks	AO	Guidance
15	(d)		**LH region** 0.20936 is (fairly) small **oe isw**	**B1**	**3.4**	allow eg r_s is closer to 0 than to 1; eg low value of r_s; eg allow r_s close to 0
			0.41998 is not close to 0 **oe isw**	**M1**	**3.4**	allow eg p-value > 0.05 (or 0.01 or 0.1) eg p-value not small do **not** allow eg p-value (relatively) large eg high p-value
			insufficient evidence to suggest association (between mean time spent running and resting pulse rate) **oe**	**A1**	**2.2b**	may infer no association
			RH region –0.61349 is closer to –1 than 0 **oe isw**	**B1**	**3.4**	allow eg r_s close to −1 eg r_s is closer to –1 than 0; eg r_s is not small **B0** for eg r_s is large **oe**
			0.00031is close to 0 **oe**	**M1**	**3.4**	allow eg p-value < 0.05 (or 0.01 or 0.10) allow eg p-value is (very) small/low
			sufficient evidence to suggest (some) [negative] association (between mean time spent running and resting pulse rate) **oe**	**A1**	**2.2b**	may infer negative association
				[6]		